

AI Discussion Paper Ver1.0

Preliminary Discussion Points for Promoting the Sound Utilization
of AI in the Financial Sector



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I. Introduction

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Background

The rapid advancements in GenAI

- Widespread implementation, with the potential to greatly enhance daily life
- However, society is increasingly aware of the risks of AI, such as crime and misinformation

In the financial sector, while AI adoption is progressing, there are hesitations due to risks and regulatory concerns

Awareness

AI has the potential to become a core technology, fundamentally transforming financial services and FI's business models

- FIs must be aware of the **“risk of not taking actions”**, which could hinder their ability to provide quality financial services in the long term

Policy responses

FSA strongly supports the sound utilization of AI by FIs.

- Provides safe harbors through clarifying regulatory expectations
- This Paper is a basis of developing long-term policies through dialogue with FIs.

II. Purpose and Positioning of this Paper

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Based on surveys and interviews with FIs, as well as progress in international discussions, this DP presents preliminary discussion points and the FSA's policy directions.

Note: The issues mentioned are based on preliminary analysis, and risks vary by use case and implementation. Institutions are encouraged to actively take on challenges without being overly cautious.

Inputs

Outputs

■ "Survey on AI Utilization by FIs"

- October 3 – November 15, 2024 (voluntary responses)
- Participants: 130 companies including banks, securities firms, and insurance companies

■ Interviews with FIs and IT vendors

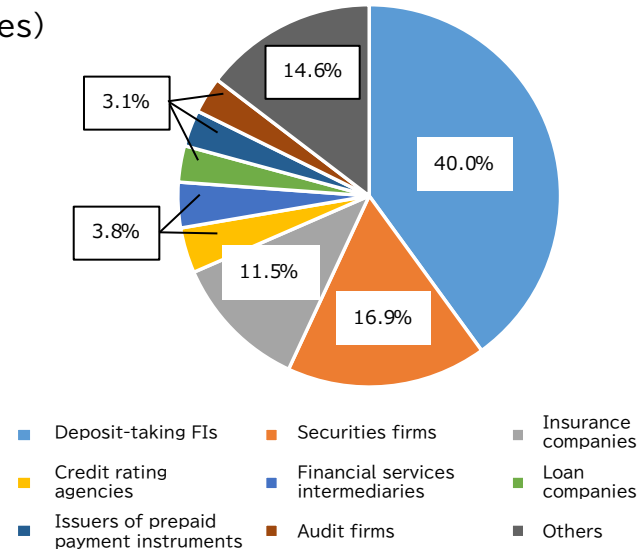
■ Progress in international discussions

■ Use cases / Challenges / Initiatives to overcome challenges by FIs (conventional AI / GenAI)

■ FSA's use of AI

■ Future direction

Chart 1: Distribution of respondent FIs by type of business

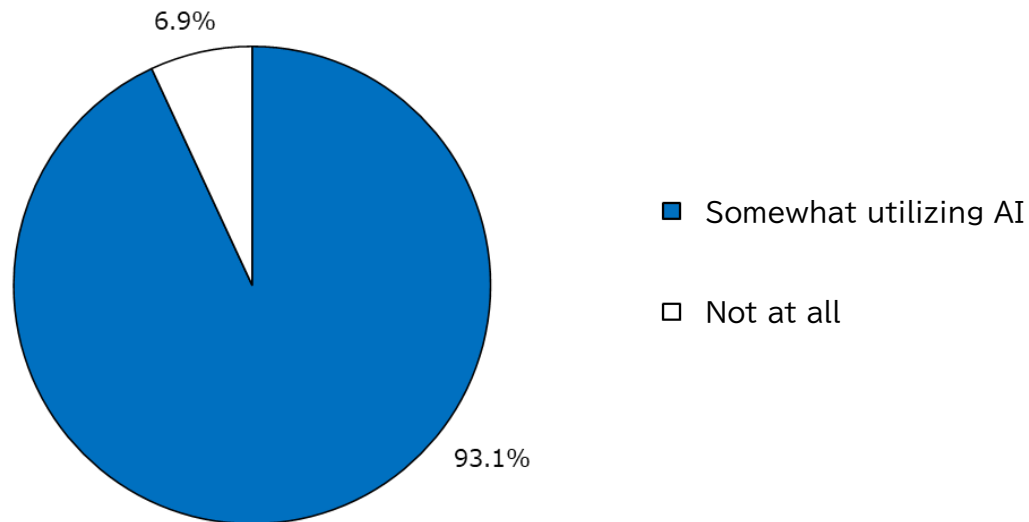


III. Use cases of AI in Finance

III. Use cases of AI in Finance

- Over 90% of respondents already use conventional AI or GenAI
- Many FIs and Fintechs have already adopted the use of AI in their operations.

Chart 2: Entities that use conventional AI or GenAI



① Improving operational efficiency

- Text to Paper (OCR)
- Information retrieval

② Use in customer service

- Chatbots
- Marketing

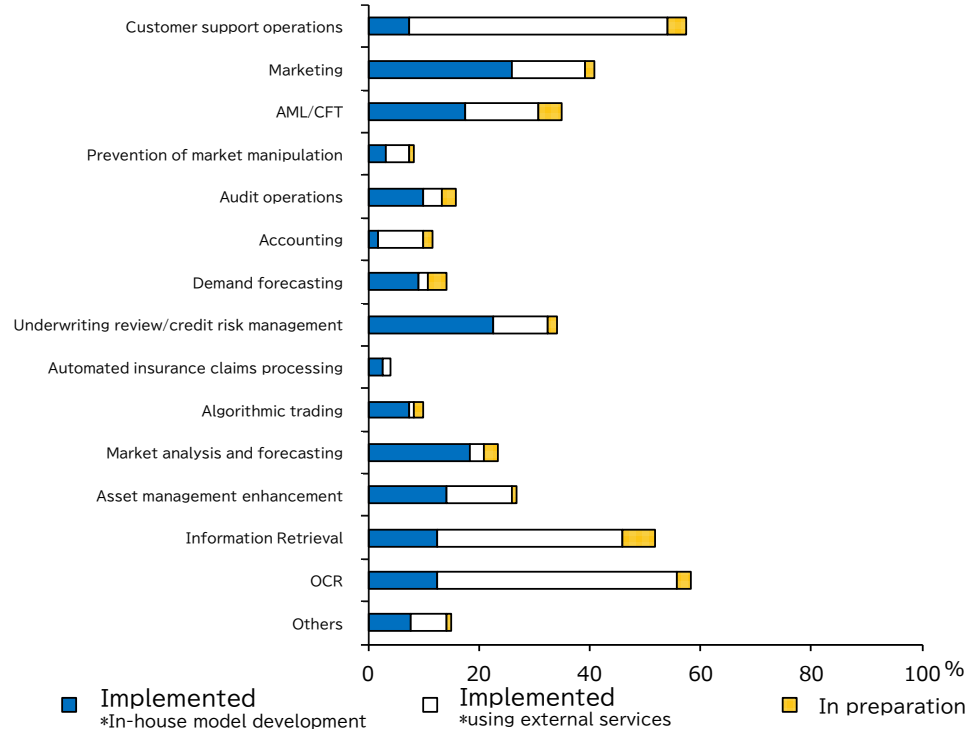
③ Enhancement of risk management

- Fraud detection (AML/CFT, etc.)
- Credit screening, credit risk management, and underwriting screening
- Sampling of Compliance Violations

④ Market forecasts, etc.

- Currency and interest rate forecasts
- Market sentiment

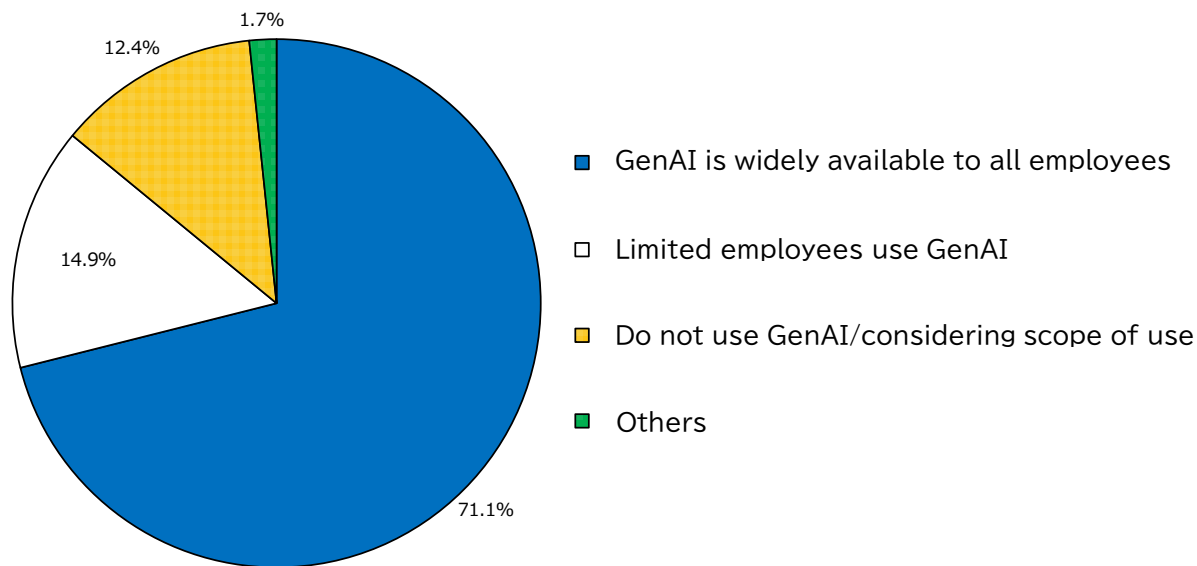
Chart 3: Use cases of conventional AI



III-2 Usage of GenAI and main use cases: Scope of use

- Approximately 70% of FIs broadly allow the use of GenAI for general employees
- Some institutions require applications for using AI, some limit usage to headquarters staff only, and others set different usage scopes depending on the type of GenAI.

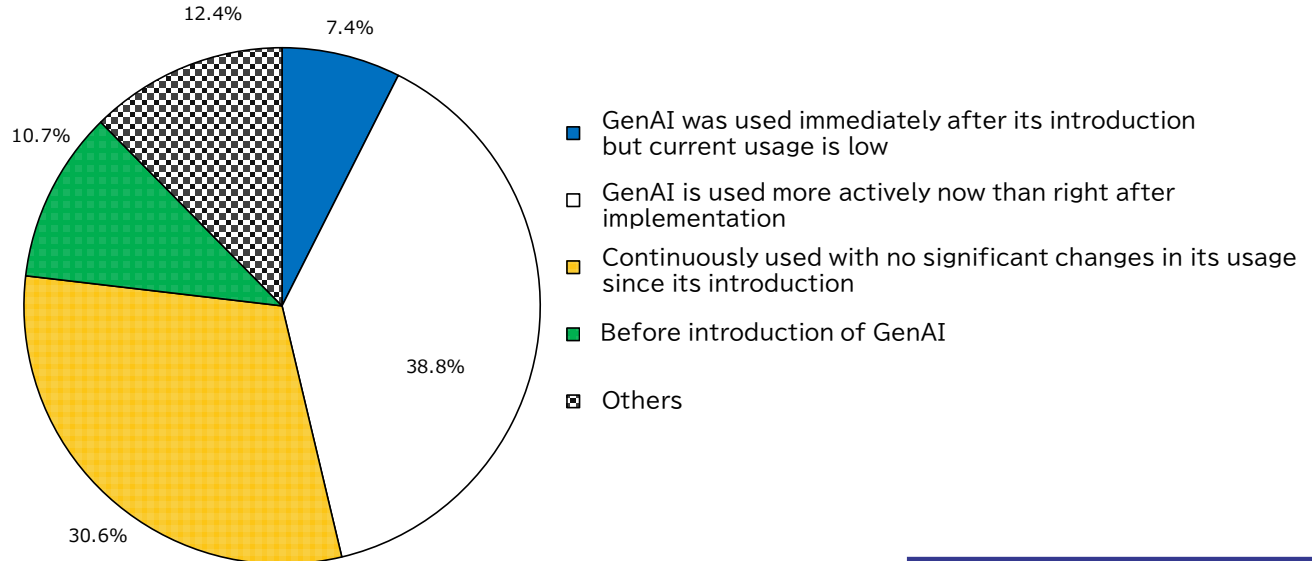
Chart 4: Scope of GenAI



III-2 Adoption of GenAI and its main use cases : after adoption

- Most FIs have either continued to use AI since it was introduced or are now using it more actively than before.
- Many institutions have only introduced general-purpose GenAI, yet its active use highlights its potential to enhance operational efficiency in financial institutions.

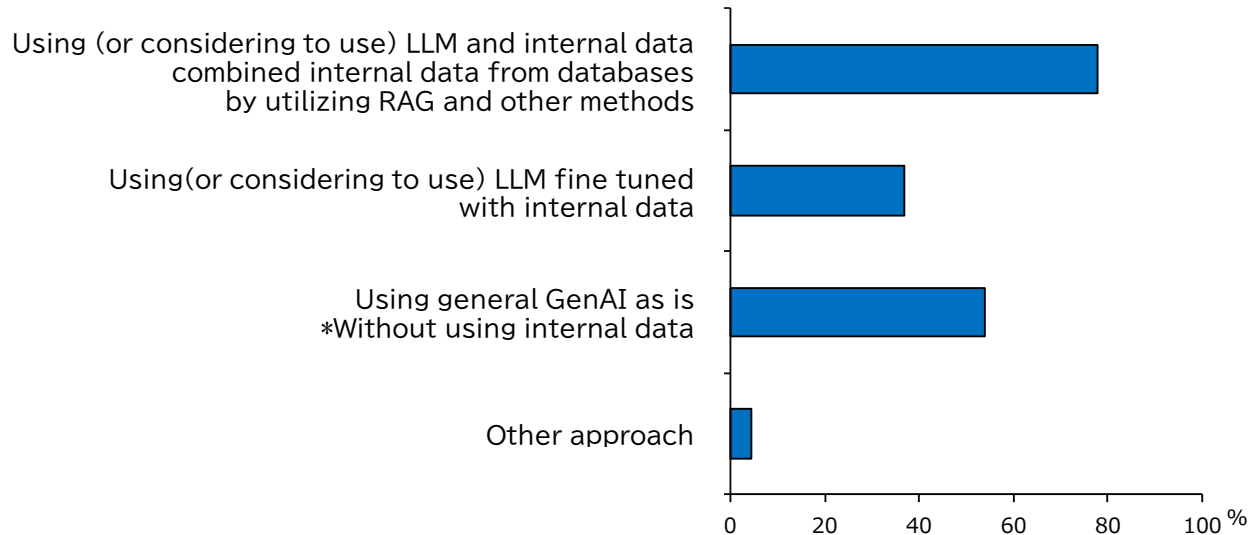
Chart 5: Usage after implementation



III-2 Current Status and main use cases of GenAI: Customization

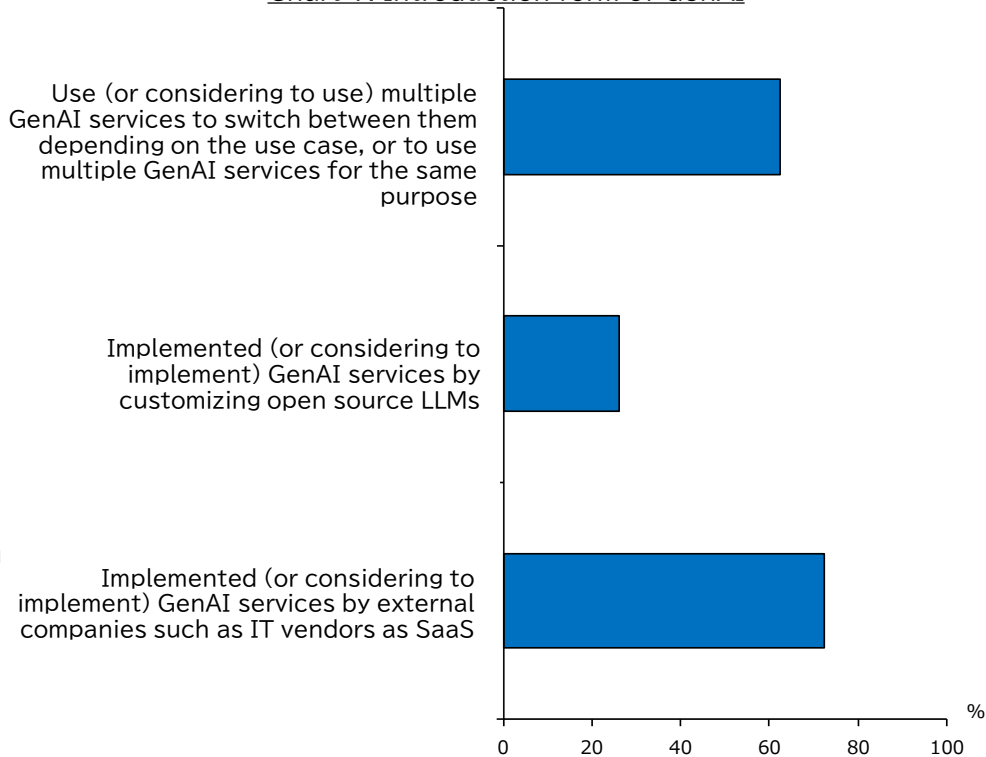
- About half of the financial institutions are using general-purpose GenAI as is.
 - Due to the ease of implementing pre-trained models, GenAI has a higher overall adoption rate compared to conventional AI.
 - On the other hand, many FIs are using or considering using LLMs provided by external IT vendors in combination with internal databases through RAG and fine tuning.

Chart 6: Introduction of GenAI (with or without customization)



- Form of GenAI implementation varies.
- Many FIs use multiple types of GenAI depending on the use case
- Some FIs use LLM which is provided as open source, in on-premise servers, while others use proprietary GenAI in closed environments via dedicated lines.
- Some have introduced GenAI tools provided by external vendors as SaaS, while others have developed their own user interfaces.

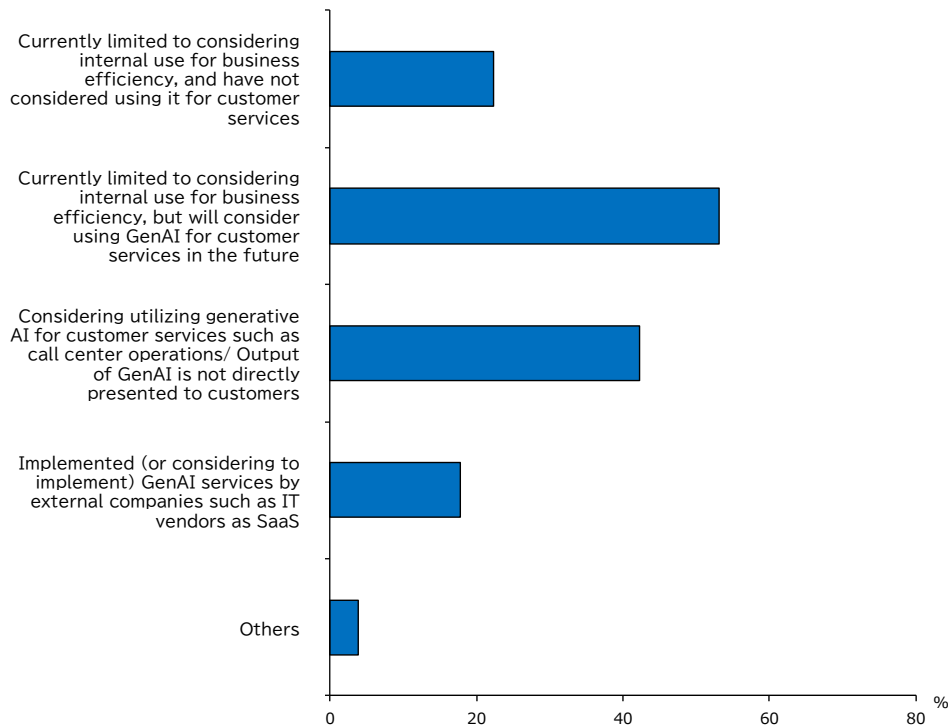
Chart 7: Introduction form of GenAI



- ① Internal use such as business efficiency improvement
- ② Indirect utilization in customer service
- ③ Direct utilization in customer service

- Currently, many FIs remain in ①.
- More than half of respondents said that they would consider using it for customer services in the future.
- ③ is very limited due to the risk of hallucination

Chart 8: Utilization of GenAI in customer service

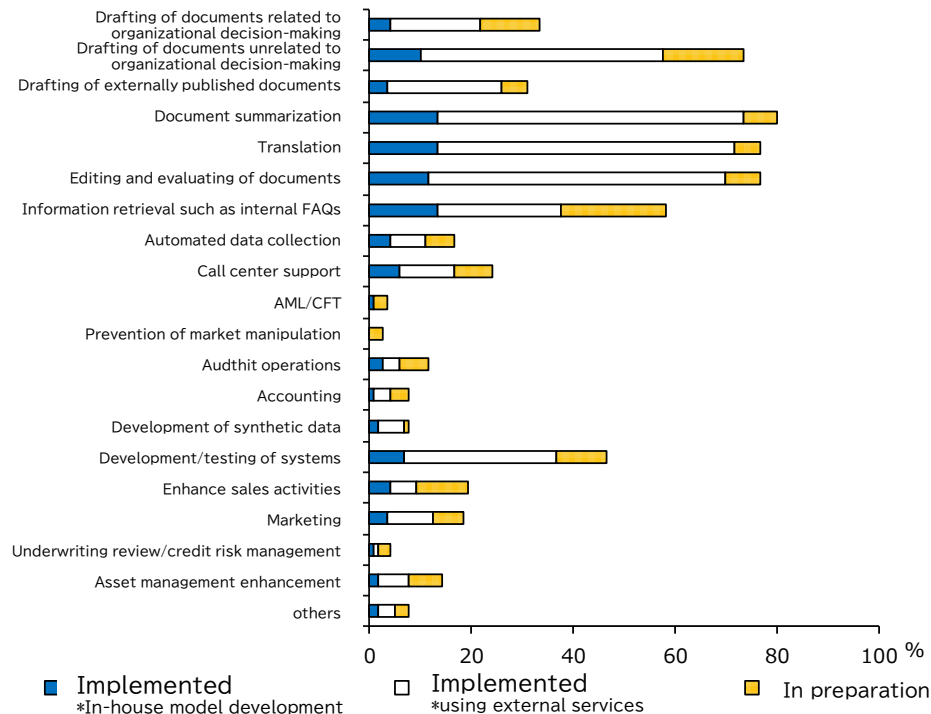


III-2 Current Status and main use cases of GenAI : Use Case Details

- ① **Internal use such as business efficiency improvement**
 - Document Summary/Translation
 - Structure, correction, and evaluation of documents
 - Information search such as internal FAQs
 - System development/testing (Non-Natural Language)
- ② **Indirect use in customer service**
 - Call center operations support
 - Preparation of draft documents such as request for approval.
 - Drafting of external announcement
- ③ **Direct use in customer service**
 - Currently very limited

- Progress in the introduction of general-purpose GenAI, such as document summarization
- Advanced use of GenAI is still in the early stage

Chart 9: Adoption of GenAI by use case



IV. Preliminary Discussion Points

① Common challenges for conventional AI and GenAI

- Data development
- Engagement with third parties and their risk management
- Return on investment

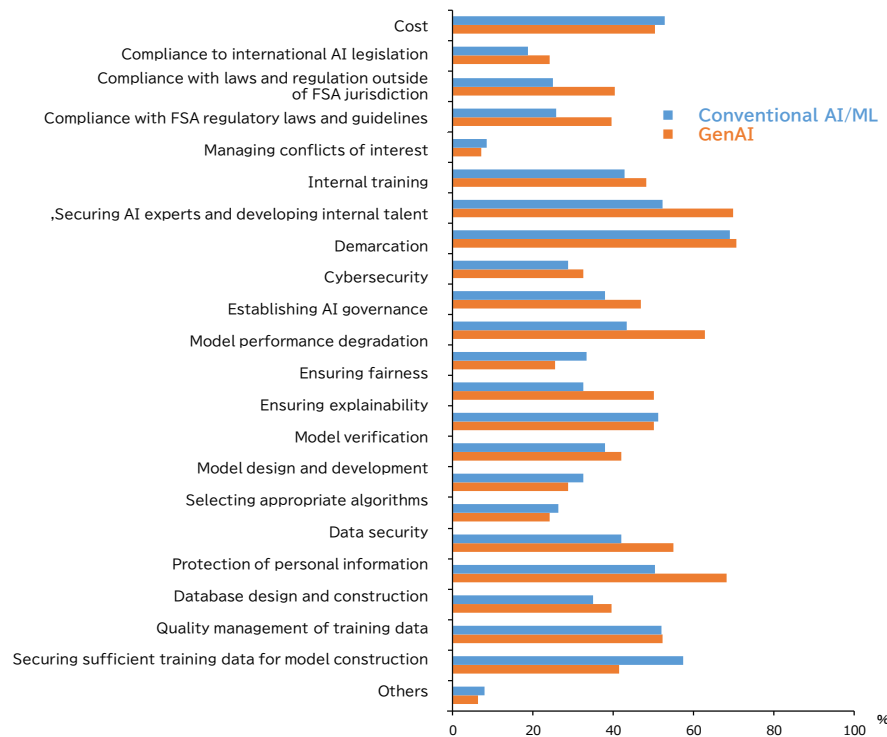
② Issues made difficult by GenAI

- Accountability
- Bias
- Model risk management
- Protection of personal information
- Information Security and Cybersecurity
- HR

③ New challenges brought by GenAI

- Hallucination
- Misuse of GenAI for financial crime
- Financial stability challenges

Chart 10: Challenges for using AI



Challenges	Examples of initiatives and future direction
■ The rapid development and widespread use of AI is making data even more important ■ Many FIs face challenges in developing internal data. ■ Around half of the FIs responded that they face challenges in (i) securing sufficient learning data for model construction and (ii) quality control of training data.	■ Progress in efforts to build databases suitable for AI use and secure training data • Build a platform for the centralized collection and management of internal data • Implementing solutions to efficiently control access to data and manage data versions • Analysis of unstructured data such as text and speech using OCR and speech recognition ■ It is crucial for FIs to recognize that their data can be an asset for increasing profitability and transforming their business model. ■ FIs should consider developing a data utilization infrastructure and establishing API connectivity as key management priorities, while ensuring security.

Challenges

- Due to the advanced expertise required for implementing complex AI models, many FIs are utilizing, or considering utilizing, solutions provided by external vendors.
- There are concerns about over-reliance on external vendors. To ensure the appropriate AI system is implemented, FIs need personnel with a certain level of AI knowledge, considering the business environment, work processes, and customer needs.
- AI-related vulnerabilities that could increase systemic risk include reliance on third parties and concentration of certain service providers (FSB report).

Examples of initiatives and future direction

- Initiatives to Address Challenges
 - Implement security checks by applying the existing third-party risk management framework
 - Leverage open-source foundation for model development to avoid over-reliance on specific vendors
 - When outsourcing the development of an AI model by providing internal data to external vendors, FIs should ensure that the IP rights of the model belong to the FIs.
- Partnerships with external vendors are essential for AI utilization, and it is necessary to select appropriate partners based on the needs of the FIs while addressing the risks associated with partnerships.

Challenges

- Benefits of using AI are difficult to estimate, making it difficult to explain the ROI, and it takes time to reach an internal agreement.
- For GenAI provided by external vendors, usage-based pricing are applied in most cases, so the cost could increase boundlessly depending on the frequency of use.

Examples of initiatives and future direction

- Initiatives to Address Challenges
 - When ROI is difficult to see and each department has difficulty in securing budget, the budget of the digital department could be used for such cases.
 - Group-wide project management to prevent multiple departments from making similar investments
- Until profitability is in sight, setting KPIs or quantitative indicators according to characteristics and monitoring against the plan is an option.
- Investment decisions should be made strategically with the appropriate understanding and proactive involvement of management.

Challenges

- GenAI has a complex learning process using a large number of parameters and a mechanism that dynamically generates output in response to multi-modal prompts such as text, audio, and images, making it difficult to clearly indicate why each inference was answered.
- A potential issue arises when sales branches are provided with AI-recommended client lists without a clear explanation of the underlying rationale, leading to underutilization.

Examples of initiatives and future direction

- It is not easy to ensure explainability in complex AI models such as GenAI.
- Progress has been made to some extent in addressing the challenges.
 - Utilizing AI model evaluation tools from external vendors.
 - To ensure user acceptance, adding the rationale behind the recommendations to the list of potential clients suggested by AI is crucial.
- It is difficult to ensure full accountability. Depending on the use case, ensuring the sense of understanding from relevant stakeholders, such as customers and employees are critical.

Challenges

- Bias in AI learning data, algorithms, and even model inference results increases the risk of unfair treatment of customers and employees with certain attributes
- Risk that the model may strengthen bias at the data training stage, for example, when training data is insufficient

Examples of initiatives and future direction

- Use of third party bias detection tools
- Given the wide variety of tools available and the lack of clear procedures for establishing specific evaluation metrics and audit processes, there is a need to establish a consistent framework for early detection and mitigation of biases in AI models

Challenges

- Even for those that responded that they have somewhat developed risk management frameworks for AI, development, implementation and management of GenAI, for which systematic performance and evaluation metrics have not yet been established, is still in its early stages.
- FIs are aware that the unique properties of GenAI, including its broad applicability and frequent model updates, make conventional risk management approaches insufficient for fully understanding and controlling all potential risks.

Examples of initiatives and future direction

- Examples of Initiatives to Address Challenges
 - Continuously strengthen existing policies and procedures for conventional AI to address GenAI
 - Conduct proof-of-concept testing to tools from multiple vendors to validate accuracy of AI models
 - Document AI model methodology, including assumptions, limits, and uncertainties.
 - Pre-production validation and ongoing monitoring of model performance by independent reviewers
 - Added feature value based on business knowledge to maintain model performance
- In the monitoring by the FSA, we will continue to engage in dialogue with FIs regarding the appropriateness of their risk management practices, considering developments in discussions on AI both domestically and international, as well as the actual usage of AI.

Challenges

The application of the rules are not clear when...

- handling personal information including the company's customer information as learning data or inputting prompts using personal information including customer information when using GenAI
- outsourcing AI model development and learning process to external vendors
- using a GenAI platform with a server located overseas

Examples of initiatives and future direction

- Examples of initiatives to Address Challenges
 - To ensure that personal information is not used as an input to the generated AI model, all employees are warned through internal administrative notifications, including notifications from the Personal Information Protection Commission.
 - Consider automatic system-based restrictions on entering customer information
 - Measures to prevent input data from being used for re-learning (establishment of dedicated sections, conclusion of contracts with vendors, etc.)
- There is a need to strike a balance between appropriate measures concerning the protection of personal information and the effective use of data. The FSA will monitor the initiatives of FIs through ongoing dialogue and consider the appropriate approach.

Challenges

- The potential for AI, including GenAI, to enhance an attacker's capabilities and increase the likelihood and impact of a cyber attack on the financial sector.
- Risk of unintended external leakage of customer information and important business information when financial institutions use GenAI
- Attacks such as prompt injection could cause AI systems to malfunction or confidential information to be leaked.
- The AI model itself may be subject to attacks, such as data poisoning.

Examples of initiatives and future direction

- Examples of Initiatives to Address Challenges
 - Set certain constraints on the input data for prompts
 - Build a system environment that can manage input / output information by introducing a GenAI service in the company's proprietary environment
 - Verification is being conducted at the planning and development stage. During the operation phase, a system is being constructed that constantly monitors output data with both human eyes and the system.
- Due to the wide range of security issues involved in the introduction and operation of AI, an organizational response such as continuous monitoring of AI model-specific vulnerabilities is essential.

Challenges

- Many FIs recognize the need to secure and train experts in AI model development, operation, and management.
- There is a concern that using external vendors may hinder the accumulation of know-how and intellectual property within the organization, and slow down internal human resource development.
- There is a lack of personnel who can select the best solutions for ongoing operation and management, and those who can bridge the gap between the business and the IT department.
- Conventional educational content is insufficient due to additional risks specific to GenAI, such as hallucination and piracy.

Examples of initiatives and future direction

- Examples of Initiatives to Address Challenges
 - Develop internal human resources through OJT while seeking support for AI model development from part-time professionals
 - Risk-taking of highly skilled internal personnel, such as SE graduates, and expansion of recruitment of new graduate experts
 - Implementation of educational measures such as support for acquisition of AI-related qualifications
 - Sharing practical examples of AI use for call center staff
 - Implemented an education program for all employees on the risks of GenAI.
 - Establishment of joint ventures with startups for in-house AI development and use
- Organizations in which multiple FIs participate are making efforts to develop human resources, and cross-industry efforts are also important.

Challenges

- A risk that the credibility of FIs may be undermined if incorrect information is presented or if responses that are directly linked to credit risk or legal risk are mistakenly generated.
- Even when RAG is used, if the business design is not appropriate, the selection of the reference sources will be insufficient and the accuracy will not be improved, and there is a risk that it will be operated without eliminating false outputs.
- The risk of employees inadvertently conveying misinformation when using GenAI to respond externally.

Examples of initiatives and future direction

- Examples of Initiatives to Address Challenges
 - Building business processes based on human involvement (Human in the loop)
 - Incorporating RAGs and prompts so that supporting documents can be included in responses to confirm supporting evidence
- It may not be appropriate to impose high accuracy requirements on AI, such as "AI must not make mistakes." Therefore, FSA's expectation is that GenAI will be actively explored, while acknowledging the societal acceptance of AI, the pace of technological development and potential use cases, and avoiding undue restraint.

Challenges

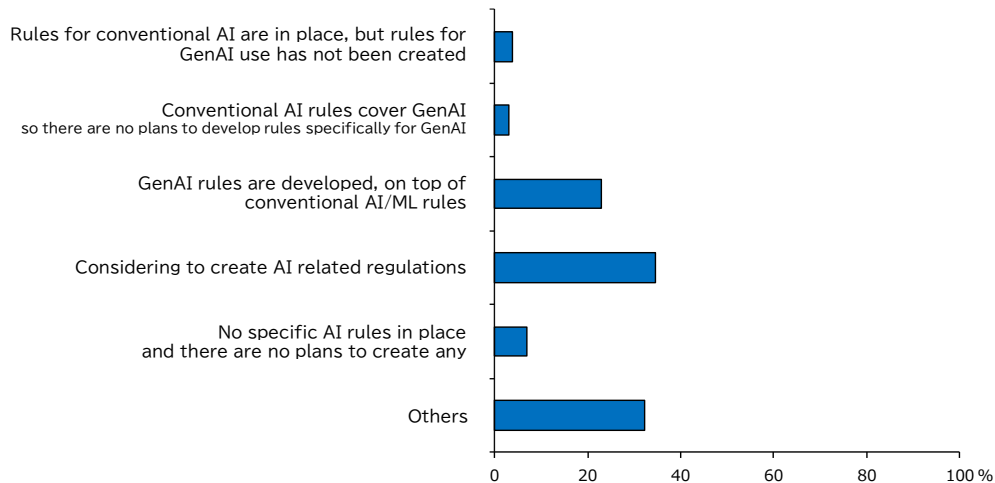
- As it has become possible to generate natural Japanese text, audio, and video using GenAI, criminal techniques have become more sophisticated, and the risks surrounding financial institutions and their customers have been expanding
- In addition to phishing scams and other scams becoming increasingly sophisticated, deepfake technology is making it easier to carry out scams through the use of fake videos and audio that impersonate specific persons.
- Conventional KYC procedures and authentication systems make it difficult to distinguish between real persons, which may lead to the possibility of identity verification being bypassed.

Examples of initiatives and future direction

- FIs are required to make efforts to identify cases of abuse of generated AI and take actions in accordance with the risks posed by generated AI, such as enhancing authentication for non-face-to-face transactions
- With regard to the monitoring by the FSA, we will identify cases of misuse of generated AI for financial crimes in cooperation with relevant ministries and agencies, and take necessary actions, such as issuing alerts to users and FIs.

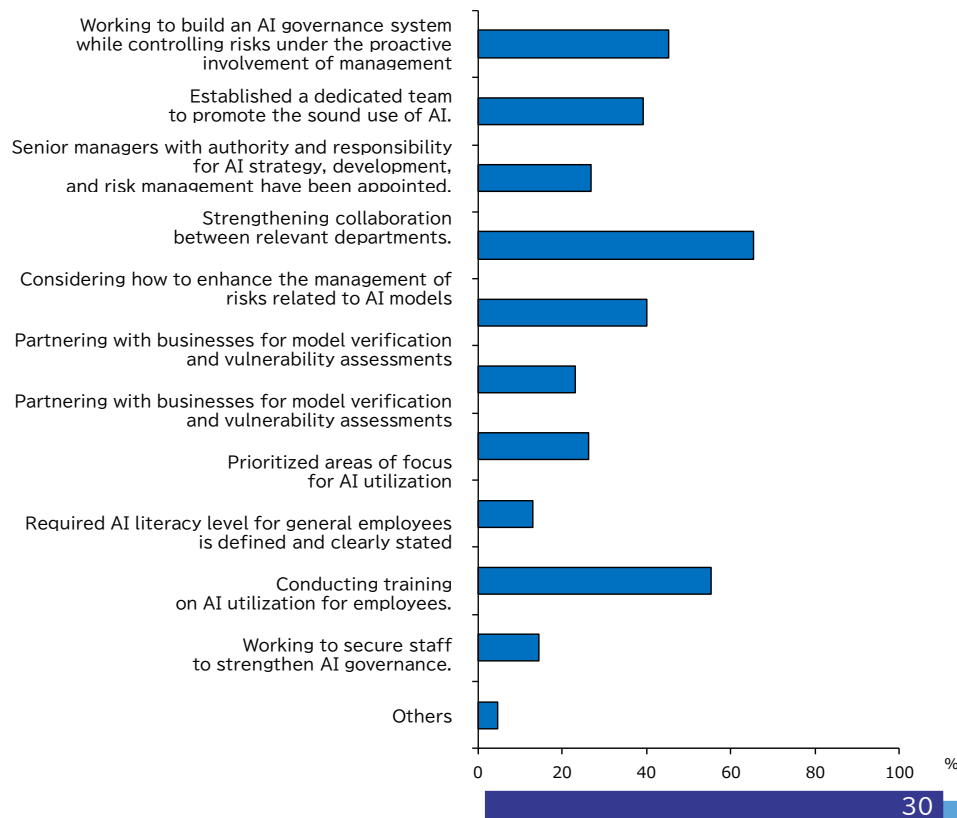
- Some FIs formulated and published AI governance guidelines and are making efforts to ensure thorough implementation at affiliated companies.
- Some FIs established internal rules for GenAI and then expanded to cover other types of AI, including conventional AI.
- There are some who responded that "the existing framework can be generally applied."
- Documents used as reference in the development of regulations: AI business guidelines, guidelines on generated AI by private organizations, FISC publications, overseas AI related laws and regulations such as the ones in EU.

Chart 11: Development of rules concerning AI



- Promotion of AI and establishment of management systems through strengthening collaboration among related departments
- Multiple departments, including the risk management department, collaborate to assess the status of risk measures with the department deploying AI.
- To prepare for the evolution of AI technologies and their growing adoption across financial institutions, define AI risk evaluation criteria from a comprehensive viewpoint.
- Due to the challenges specific to AI, currently, many FIs are considering sound AI utilization through trial and error while applying the conventional governance and risk management framework.

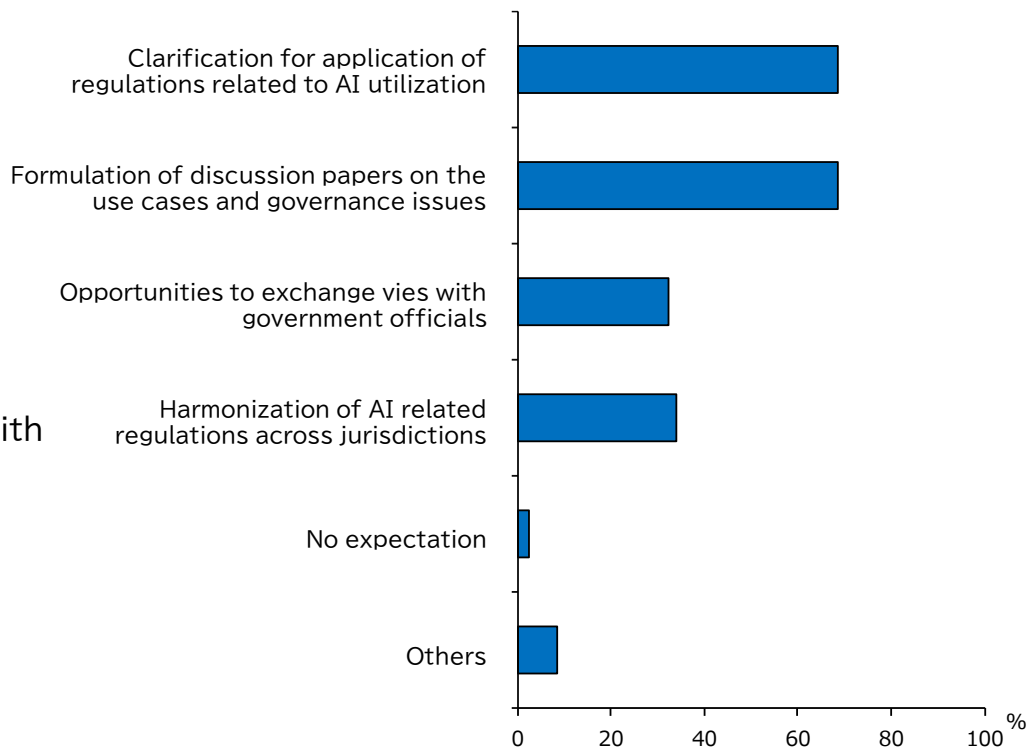
Chart 12: Initiatives towards building AI governance



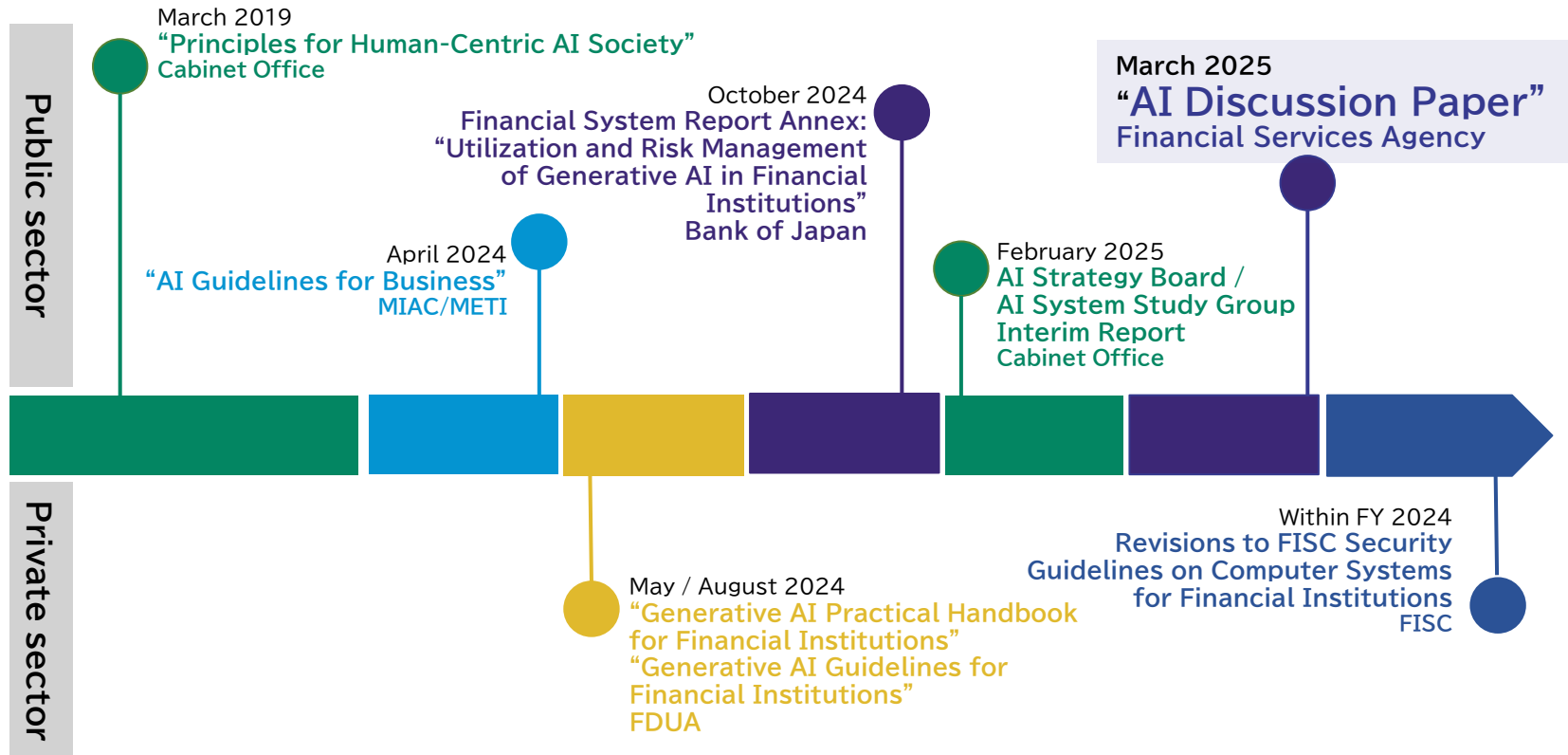
IV-3 Future direction: Expectations to the FSA

Chart 13: Main expectations to the FSA

- Clarification for application of regulations related to AI utilization
- Harmonization of AI-related regulations across jurisdictions
- Opportunities to exchange views with government officials



IV-3 Recent Actions by the Public and Private Sectors



G20 Minister of Finance and Central Bank Governors Meeting

in February 2024

The meeting stressed the importance of understanding the benefits and vulnerabilities coming from digital innovations, including artificial intelligence (AI).



Financial Stability Board November 2024

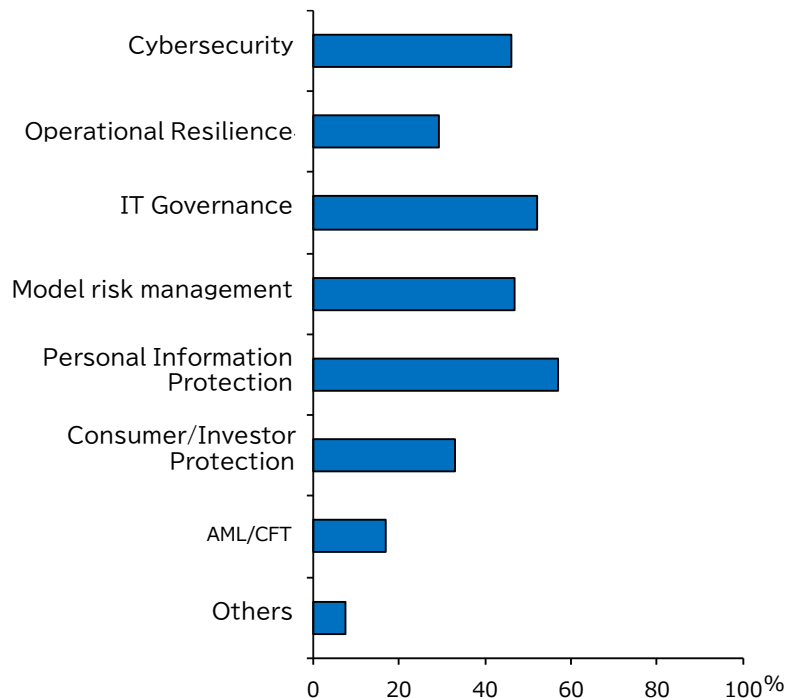
- The Financial Stability Board (FSB) published a report on the financial stability implications of AI in November 2024 for submission to G20 Leaders.
- The International Organization of Securities Commissions (IOSCO) plans to publish a report by the first quarter of 2025 that will summarize market participants' use cases for new AI technologies and the issues, risks and challenges for considering potential policy responses, with a view toward ensuring the soundness of markets and other issues.
- Progress on AI-related discussions in other SSBs (BCBS, IAIS, IFIAR, etc.)



■ Clarifying the application of regulations

Chart 14 Areas where clarifying the application of regulations is needed

- Privacy, IT governance, model and risk management, and cybersecurity are the top areas that require clarification
- For all issues, there are existing laws, regulations, supervisory guidelines, principles, and guidelines that apply regardless of whether or not AI is used. Therefore, we will encourage FIs to take actions in line with these regulations.
- While, taking into account new issues arising from the characteristics of generated AI, we will examine whether the regulatory requirements for AI utilization have been sufficiently clarified and whether the existing regulatory and supervisory frameworks have sufficiently addressed risks.
- If a significant regulatory gap is identified, consider revising the principles and guidelines first, based on the government's policy that regulatory measures should be limited to those that cannot be expected to be addressed by independent efforts of business operators.



- **“Multi-stakeholder Study Group”** will be held to deepen discussion on specific issues

■ FinTech Support Desk

- A centralized contact point for consultation on laws and regulations, etc. from businesses running businesses that involve various innovations or businesses that are considering new businesses.
- Since its establishment in 2015, the Desk has received a total of 2380 consultations from FIs and Fintechs as of December 2024. A framework that can be used when seeking clarification of the interpretation of laws and regulations concerning the use of GenAI

■ FinTech Proof-of-Concept Hub

- A framework to support proof-of-concept testing for innovation by forming a team within the FSA and collaborating with other ministries and agencies as necessary.
- AI-related selected projects include a proof-of-concept trial using AI to improve the efficiency of financial institutions' compliance operations report published in August 2018



FinTech Support Desk
Website

Multi-stakeholder Study Group

March 2025

June to December 2025 *TBD

March 2026

Publication of AI Discussion Paper (Ver.1.0)

Japan Fintech Week 2025

Updating the Paper

Japan Fintech Week 2026

Domestic Response

Providing consultations (FinTech Support Desk, PoC Hub, etc.)

Monitoring and environmental development for discussion points in this DP
(including review of guidelines)

International Response

Contributions to international discussions (FSB, IOSCO, BCBS, IFIAR, etc.),
dialogues with national authorities

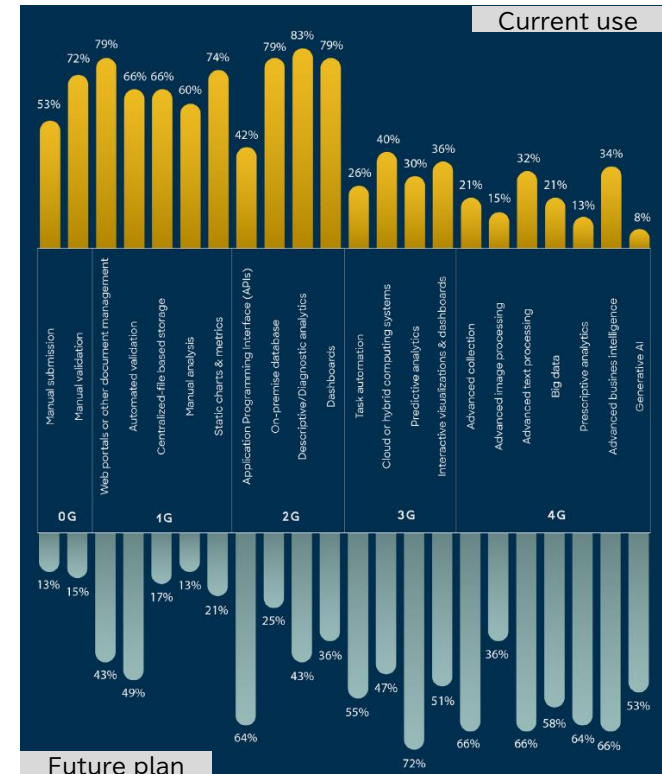
Gathering information e.g., Participations to fintech conferences

V. FSA's use of AI

V. FSA's use of AI

- According to a survey of 64 financial supervisory authorities conducted by the CCAF, 59% of those subject to the survey have implemented at least one Suptech application
- AI-related technologies (e.g., text processing, big data analysis) have been used in Suptech only to a limited extent, but many authorities have expressed interest in using them in the future.
- Examples of Initiatives by National Authorities
 1. Prediction of financial crisis using big data (Bank of England)
 2. Extraction of patterns from unstructured data (e.g. corporate information) (UK PRA)
 3. Use of AI in market surveillance (AMF, France)
 4. Examples of Initiatives at the FSA
 - Risk analysis using granular data (prediction of the credit risk (credit rating of a borrower) for real estate loans-using a machine learning technique, analysis of the impact of HFT on the magnitude of market fluctuations using a neural network)]
 - Efficient extraction of financial institution information through text analysis

Chart 13: SupTech Adoption



Source: Cambridge Centre for Alternative Finance

VI. Conclusion

■ Importance of multi-stakeholder approach

- As the use of AI is accelerating, strengthening collaboration across the financial industry and society as a whole is significant in order to overcome challenges and to maximize the benefits of AI.
- Public-private partnership initiatives could be effective to create use cases and build AI governance, as small FIs have limited capacity.
- FSA's collaboration with other ministries and agencies and contribution to international rule making are important.
- **FSA intends to support sound AI utilization in the financial sector**
- FSA is determined to continue providing opportunities for dialogue between public and private stakeholders, both domestic and abroad, at opportunities such as Japan Fintech Week, and make every effort to realize open innovation while improving the predictability of regulatory supervision.
- FSA plans to host **Multi-stakeholder Study Group** on AI.

■ Request for comments on the DP

- The analysis in this DP is preliminary, and the issues raised are likely to change significantly as a result of technological innovations and changes in the business environment.
- Please contact Fintech and Innovation Office, Risk Analysis Division, Strategy Development and Management Bureau, for any comments or suggestions.

Contact us

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Financial Services Agency

